HOCHSCHULE
HAMM-LIPPSTADT

EXPERIMENT 2: TRANSISTOR AMPLIFIERS CHARACTERISTICS AND APPLICATIONS

Subject: Electronic Engineering III
Hamm-Lippstadt University of Applied Sciences, Lippstadt
Location: L3.3-E02-080

Scheduled day for the experiment: 08.01.2025 and 09.01.2025
The pre-report must be submitted before conducting the experiment
The final report is due on **January 16, 2025.**

Student Assistance

Office hours should be scheduled with the professor via email, providing the subject of the question in advance.

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Experiment Objectives

- Explore the fundamental operating principles of bipolar junction transistors (BJTs) in amplifier circuits.
- Analyze the role of transistors in signal amplification, focusing on the common emitter and common collector configurations.
- Investigate the voltage gain, phase shift, and current amplification properties enabled by BJTs.
- Develop practical skills in assembling and testing transistor-based amplifier circuits.
- Enhance familiarity with laboratory instruments while characterizing transistor performance in different configurations.

I. INTRODUCTION

Transistors are essential components in electronic engineering, widely utilized for signal amplification and switching applications. This experiment focuses on understanding the operational principles of bipolar junction transistors (BJTs) and their behavior in amplifier configurations.

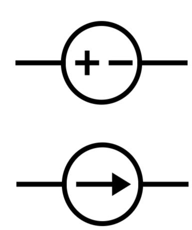
The components and devices used in this experiment, including transistors, independent voltage sources, resistors, capacitors, multimeters, and breadboards, are detailed in Section II. Section III outlines the two primary tasks of this experiment:

1) **Task 1:** Assembly and analysis of a single-stage common emitter amplifier circuit to study its amplification characteristics, including voltage gain and phase inversion.
2) **Task 2:** Assembly and analysis of a single-stage common collector amplifier circuit to investigate its current gain properties and input/output impedance behavior.

II. COMPONENTS AND DEVICES

- **Independent Voltage Source and Independent Current Source**
Independent Voltage Source: A device designed to maintain a fixed voltage across its terminals, regardless of the current flowing through it or the circuit elements connected to it. A common example of this device is a battery.
Independent Current Source: A device that ensures a specified current flows through itself, independent of the voltage across its terminals or the connected circuit elements. Unlike voltage sources, practical examples of current sources are less common; however, they are valuable for constructing theoretical models. A multimeter can function as both a current and a voltage source in laboratory setups. Table I illustrates the symbols and device examples for independent voltage and current sources.

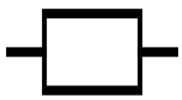
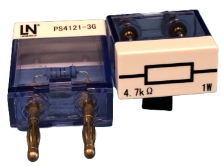
TABLE I: Symbols and devices for independent voltage and current sources.

Symbols	Device Photo
	

- **Resistor**
Resistors are passive electrical components that resist the flow of current, dissipating energy as heat. The resistance values of the THT resistors used in

this experiment are indicated by the color bands on their surface. Table II presents the symbol and a photograph of the resistors.

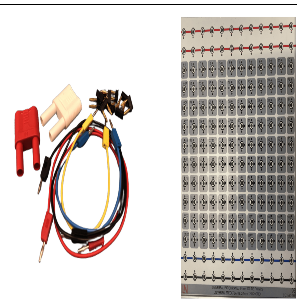
TABLE II: Resistor symbol and device photo

Symbol	Device Photo
	

To correctly measure resistance, the equipment must be properly connected. When using a multimeter as an ohmmeter, connect it across the two terminals of the resistor. Ensure that the circuit is powered off during this measurement to avoid inaccuracies or damage to the multimeter. For voltage measurements, a voltmeter should always be connected in parallel with the component. Conversely, for current measurements, an ammeter must be connected in series with the component to accurately measure the current flowing through it.

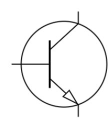
- **Conductor (Wires and Breadboard)**
  A conductor is a material that allows charge to flow through it. It is used to connect circuit elements in an electrical circuit. The conductors commonly are made of metals, as they have low resistance and high conductance. Breadboard, which is a board widely used as a prototype, acts as a conductor and the construction base of electronics. The breadboard allows the quick creation of circuits without making permanent connections. It is a widely used tool to learn electrical circuits. Table III shows the symbol and the photo of the wires and breadboard. Table III shows the symbol and the photo of the wires and breadboard.

TABLE III: Conductor symbol and device photo

Symbol	Device Photo
	

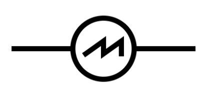
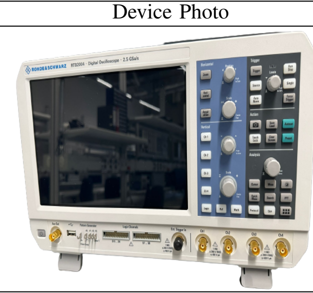
- **Transistor**
  A transistor is a semiconductor device designed to amplify or switch electrical signals and power. It is one of the fundamental building blocks of modern electronics. It is composed of semiconductor material, usually with at least three terminals for connection to an electronic circuit. A voltage or current applied to one pair of the transistor's terminals controls the current through another pair of terminals. Because the controlled (output) power can be higher than the controlling (input) power, a transistor can amplify a signal. Transistors can be packaged as discrete components or integrated into complex circuits within integrated circuits. Table IV presents the symbol and a photograph of a typical transistor.

TABLE IV: Transistor symbol and device photo

Symbol	Device Photo
	

- **Oscilloscope**
  The oscilloscope is an electronic test instrument that graphically displays voltage variations over time. It allows analysis of waveform properties, including amplitude, rise time, and period. Table V presents the symbol and a photograph of the oscilloscope.

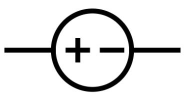
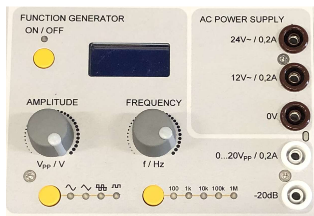
TABLE V: Oscilloscope symbol and device photo

Symbol	Device Photo
	

- **Function Generator**
  A function generator produces standard waveforms such as sine, square, and triangular waves. The instrument allows precise control over parameters such as amplitude and frequency, making it essential to test and analyze electronic circuits. Table VI

shows the symbol and a photograph of the function generator.

TABLE VI: Function Generator symbol and device photo

Symbol	Device Photo
	

III. Experiments

This section describes the experiments conducted to analyze the characteristics and applications of BJT amplifiers. The tasks focus on two fundamental amplifier configurations, emphasizing their unique properties and practical relevance in analog circuits. The experiments include:

- **Task 1:** Analysis of the Single Stage Common Emitter Amplifier, focusing on voltage amplification and phase inversion.
- **Task 2:** Investigation of the Single Stage Common Collector Amplifier, highlighting its current buffering capabilities and impedance matching properties.

A. Task 1: Single-Stage Common Emitter Amplifier

The Single-Stage Common Emitter Amplifier is a circuit that utilizes a bipolar junction transistor (BJT) with the emitter as the common terminal. It is designed to amplify input signals by modulating the collector current based on variations in the base-emitter junction caused by the input signal. This configuration provides moderate to high voltage gain and outputs a signal that is 180º out of phase with the input. Coupling capacitors are employed to block DC components and ensure only the amplified AC signal is transmitted.

The BD237 is a medium-power NPN transistor, chosen for its robustness and high current capability, making it well-suited for the experimental setup described. Its characteristics ensure reliable performance in this amplification circuit.

The schematic diagram of the circuit is shown in Figure 1.

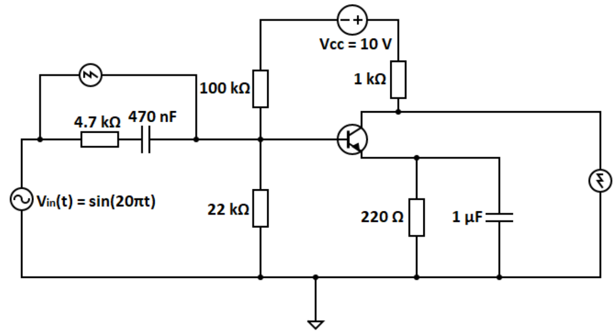


Fig. 1: Single-Stage Common Emitter Amplifier Circuit.

Procedure:

1) **Assemble the Circuit:** Using the components available, construct the Single-Stage Common Emitter Amplifier as illustrated in Figures 1 and 2.

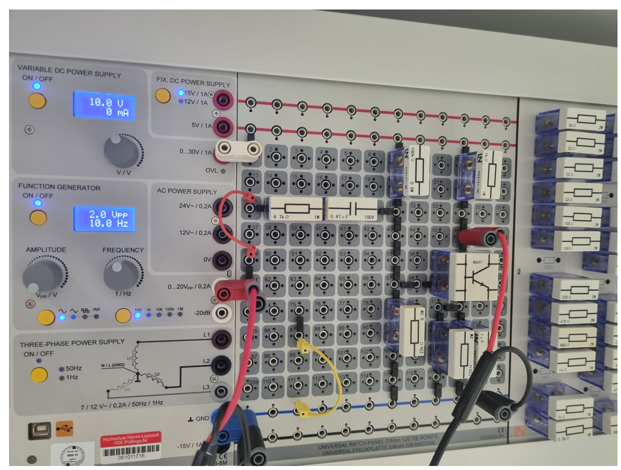


Fig. 2: Assembled Single-Stage Common Emitter Circuit.

2) **Configure the Function Generator and Oscilloscope:** Set the function generator to produce a sinusoidal input signal (V_{in}) with the following parameters:
   - Amplitude: $V_{pp} = 2\,\text{V}$ (peak-to-peak voltage)
   - Frequency: $f = 10\,\text{Hz}$

   Connect the output of the function generator to the circuit input (V_{in}). Configure the oscilloscope as shown in Figure 3, with the following channel settings:
   - **Channel 1**: Connect the probe to the circuit input (V_{in}), ensuring the ground clip is attached to the circuit's ground.
   - **Channel 2**: Connect the probe to the circuit output (V_{out}), ensuring proper grounding.
3) **Observe the Waveforms:**

Equipment

The following equipment is needed for this experiment and should be configured as shown:

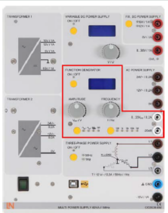
Equipment	Settings	Channel A	Channel B
	Sensitivity	1 V/DIV	1 V/DIV
	Coupling	DC	DC
	Polarity	norm	inv
	Y position	0	0
	Time base	4 ms/DIV	
	Mode	X/T	
	Trigger channel	A	
	Trigger edge	pos	
	Waveform	sine	
	Amplitude	1V	
	Frequency factor	1	
	Frequency	10 Hz	

Fig. 3: Oscilloscope and Function Generator Settings for the Task 1 of the Experiment 2.

- On the oscilloscope, observe both the input and output signals.
- Measure the peak-to-peak voltage (V_{pp}) of V_{in} to validate the input signal configuration, and then focus on V_{out} to analyze the amplification and phase inversion.
- Verify if the output signal is the amplified and inverted version of the input signal (180º out of phase with the input signal).

4) **Document the Results:**
- Take clear photographs of the oscilloscope display showing both input and output signals for the first connection setup.

5) **Repeat the Measurements:**
- Change the oscilloscope connection to observe the output signal at the right side of the circuit (after the coupling capacitor), as shown in Figure 1. Remember to ensure proper grounding after reconnection.
- Repeat the process: observe the waveforms, measure the peak-to-peak voltages (V_{pp}), and verify that the output signal is amplified and inverted (180º out of phase with the input signal). Record these measurements.

6) **Document the Results for the Second Configuration:**
- Again, take photographs of the oscilloscope display showing both V_{in} and V_{out} signals under the second configuration.

7) **Analyze and Discuss the Results:**
- Calculate the voltage gain (A_v) of the amplifier using the formula:

$$A_v = \frac{V_{out}}{V_{in}}$$

- Discuss any discrepancies between the theoretical expectations and the experimental results, considering potential factors such as component tolerances, measurement uncertainties, and non-idealities in the circuit or equipment.

B. *Task 2: Single Stage Common Collector Amplifier*

The Single Stage Common Collector Amplifier is a circuit that also utilizes a bipolar junction transistor (BJT). In this configuration, the input signal is applied to the base terminal, and the output signal is taken from the emitter terminal. The collector terminal is common to both input and output circuits, hence the name. The collector terminal is effectively "grounded" or connected to the power supply.

The common collector amplifier, often referred to as an emitter follower, is particularly useful for its high input impedance and low output impedance characteristics. This configuration is well-suited for buffering applications, as it provides current gain while maintaining a voltage gain close to unity.

Figure 4 illustrates the circuit schematic.

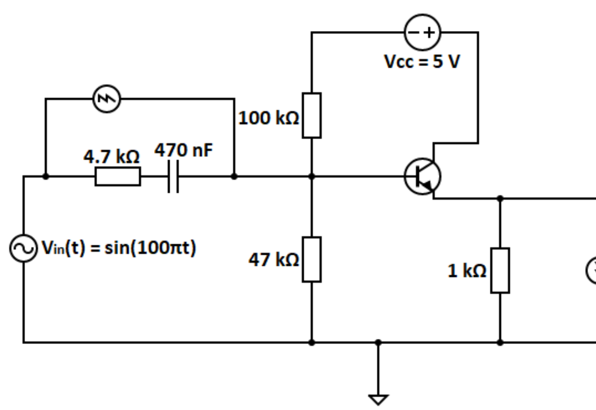


Fig. 4: Single Stage Common Collector Amplifier Circuit.

Procedure:

1) **Assemble the Circuit:** Using the components provided, construct the Single Stage Common Collector Amplifier as shown in Figures 4 and 5. Double-check all connections and ensure the proper orientation of components, particularly the transistor.
2) **Configure the Function Generator and Oscilloscope:** Set the function generator to produce a

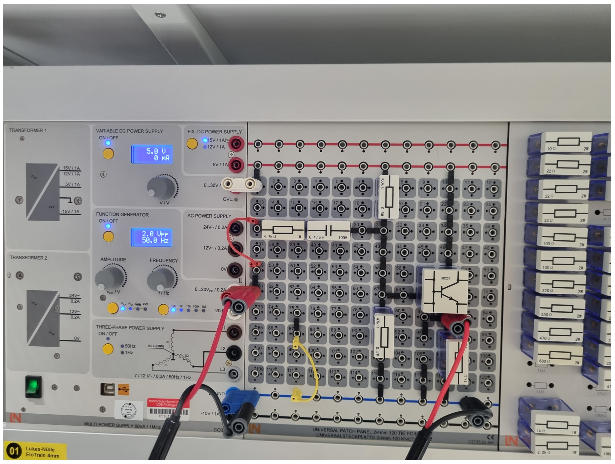


Fig. 5: Assembled Single Stage Common Collector Circuit.

sinusoidal input signal (V_{in}) with the following parameters:

- Amplitude: $V_{pp} = 2\,\mathrm{V}$ (peak-to-peak voltage)
- Frequency: $f = 50\,\mathrm{Hz}$

Connect the output of the function generator to the circuit input (V_{in}). Configure the oscilloscope channels similarly to Figure 3, noting that the frequency is 50 Hz:

- **Channel 1**: Connect the probe to the circuit input (V_{in}), ensuring the ground clip is attached to the circuit's ground.
- **Channel 2**: Connect the probe to the circuit output (V_{out}), ensuring proper grounding.

3) **Observe the Waveforms (First Configuration):**
- On the oscilloscope, observe both the input (V_{in}) and output (V_{out}) signals for the first connection setup.
- Measure the peak-to-peak voltage (V_{pp}) of V_{in} to validate the input signal configuration, and then measure V_{out} to analyze the amplification.
- Verify that the output signal is synchronized in phase with the input signal, confirming no phase inversion.
- Take clear photographs of the oscilloscope display showing both V_{in} and V_{out} signals for the first connection setup. Record the measured peak-to-peak voltages (V_{pp}) and any observations about the phase relationship and voltage gain.

4) **Repeat the Measurements for the Second Configuration:**
- Change the oscilloscope connection to observe the output signal at the second measurement point, as shown on the right side of Figure 4. Ensure proper grounding after reconnection.
- Repeat the process: observe the waveforms, measure the peak-to-peak voltages (V_{pp}), and confirm that the output signal remains synchronized in phase with the input signal. Record these measurements.
- Take clear photographs of the oscilloscope display showing both V_{in} and V_{out} signals for the second configuration. Record the measured peak-to-peak voltages (V_{pp}) and any observations about the phase relationship and voltage gain.

5) **Analyze and Discuss the Results:**
- Confirm that the voltage gain (A_v) of the amplifier is close to unity ($A_v \approx 1$) by comparing the peak-to-peak voltages (V_{pp}) of V_{in} and V_{out}.
- Highlight the current buffering characteristic of the common collector amplifier, emphasizing its high input impedance and low output impedance.
- Reflect on any discrepancies between theoretical expectations and experimental results, considering factors such as component tolerances, measurement uncertainties, or equipment limitations.
- Summarize your findings in a written comment, focusing on the practical applications of the common collector amplifier.

Pre-Report for Electronic Engineering III - Experiment 2: Transistor Amplifiers Characteristics and Applications

Group:
Date:

Student Identification

Matriculation Number	Student Name
2219941	Md Mostafizur Rahman
2220046	Ikramul Hasan Sazid

Pre-Report Tasks

- Perform simulations for Tasks 1 and 2 using any appropriate software.
- Predict theoretical values for the parameters listed in the measurement table and fill in the corresponding column.
- The goal is to compare these theoretical predictions with the experimental results obtained during the lab session.
- Submit the pre-report before the experiment, attaching this page along with the completed simulation results.

Measurement Parameters by Task

TABLE VII: Measurement Parameters Organized by Task

Task	Parameter	Description	Measured Value
1	V_{pp} of V_{out} (1st Configuration)	Peak-to-peak voltage of the output signal measured at the first oscilloscope configuration (left side of Figure 1).	2V
1	V_{pp} of V_{out} (2nd Configuration)	Peak-to-peak voltage of the output signal measured at the second oscilloscope configuration (right side of Figure 1).	2:359
2	V_{pp} of V_{out} (1st Configuration)	Peak-to-peak voltage of the output signal measured at the first oscilloscope configuration (left side of Figure 4).	2V
2	V_{pp} of V_{out} (2nd Configuration)	Peak-to-peak voltage of the output signal measured at the second oscilloscope configuration (right side of Figure 4).	1:487

Grades for Simulations

N_1	N_2	N_F

Notes:

- N_1: Grade for Task 1 simulation.
- N_2: Grade for Task 2 simulation.
- N_F: Final grade, calculated as the average of the two tasks:

$$N_F = \frac{N_1 + N_2}{2}$$